Paper review · Visionary

Confidence calibration of fine-tuned OpenVLA

OpenVLA, Kim, Pertsch, Karamcheti et al., 2024.

arXiv:2406.09246 · June 2024
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bushuyeu.com/review-openvla-visionary

Does fine-tuning change calibration?

Monitors depend on one fine-tuned model

Help-request monitors use token confidence directly (Zollo and Zemel), through a classifier (INSIGHT), with conformal prediction (ReconVLA) or through a calibrator (VLAConf). Each monitor is fitted to one fine-tuned model.

The gap

In language models, LoRA on small datasets causes overconfidence, and Bayesian LoRA reduces it. I found no controlled test for VLAs. Zollo and Zemel measured only released OpenVLA checkpoints (LoRA, rank 32).

Proposed research

Hypotheses

H1 10% of the demonstrations gives a larger confidence-minus-success gap than 100%. At similar success, LoRA gives a larger gap than full fine-tuning.

H2 Bayesian LoRA (Laplace-LoRA, 4 adapters) gives a lower calibration error than temperature or action-wise Platt scaling.

H3 Under LIBERO-Plus camera, lighting and layout changes, the LoRA-versus-full gap increases with frozen calibrators.

If H1 holds, each new fine-tune needs new labeled rollouts. If H2 holds, a posterior over the adapter repairs the confidence without a new model.

Design

Model	OpenVLA-7B, released pretrained weights.
Fine-tuning	Full, LoRA rank 8, LoRA rank 32 (default). Same step budget.
Data	10% (sampled per task) and 100% of the demonstrations.
Confidence	Raw, temperature, action-wise Platt, Laplace-LoRA, 4-adapter ensemble. One acting adapter.
Tasks	LIBERO Spatial and Long, then Object and Goal. LIBERO-Plus for H3.
Metrics	Success, confidence minus success, ECE, Brier, NLL, failure AUROC.
Trials	500 per configuration, one seed. 20% / 80% calibration and test splits.

Compute and risks

Compute (estimate)

24 runs on Spatial and Long, 4 of them full fine-tuning. 360–780 A100-hours at OpenVLA's rates, before Laplace fitting and seeds.

Evaluation needs about 200 GPU-hours. The measured rate is 7 steps per second on one RTX 5090.

Risks

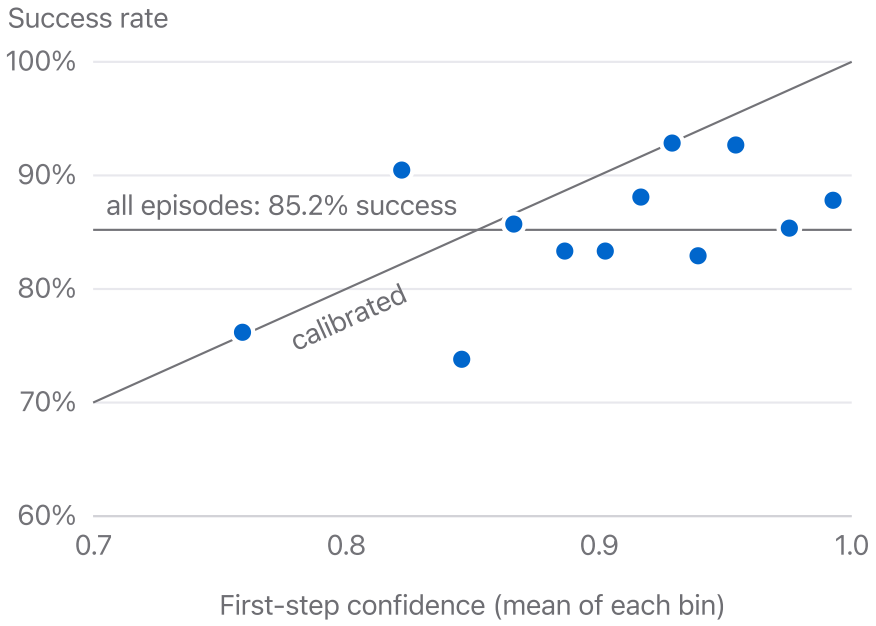
Token-based success confidence is a proxy, and it is weakest late in failed episodes.

About 75 failures in 500 Spatial episodes and 230 in Long. A 20% split holds about 15 Spatial failures. An acting ensemble costs 4× the inference.

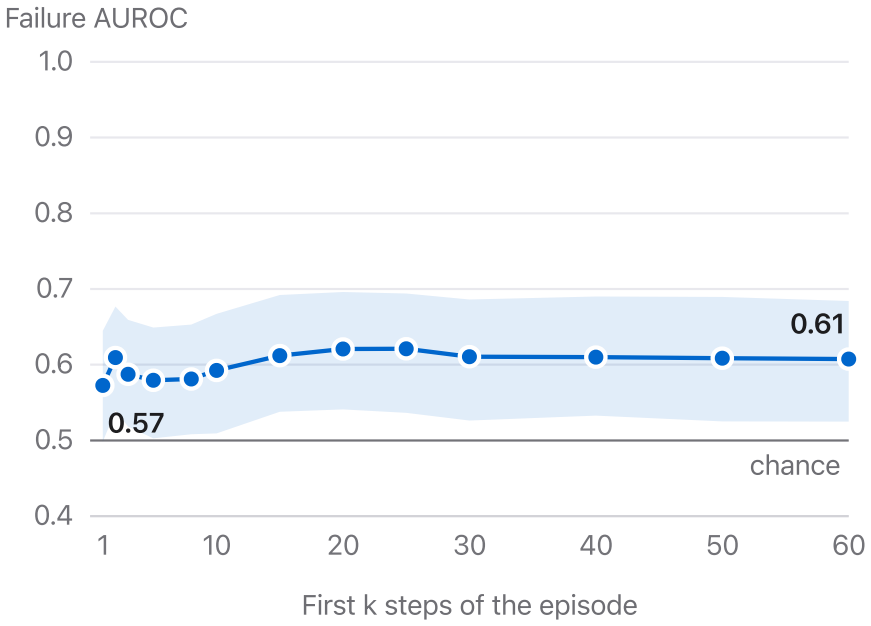
Preliminary result

Calibrated on average, weak at predicting failure

85.2% success (paper 84.7%) **0.061** ECE (study 0.088) **+4.7** points overconfident **0.57** failure AUROC



Success stays near 85% whatever the first-step confidence.



Over the first 60 steps the AUROC stays near 0.6, little better than chance.

Released LIBERO-Spatial checkpoint, 500 episodes, one seed. The whole-episode AUROC of 0.83 comes only after the successes end.

Application: rehearsal in a digital twin

1. **Scan** the workspace and its objects with 1–3 minutes of phone video.
2. **Rebuild** the scene with object meshes, the robot and a matched camera. Set masses and friction by hand, and randomize them.
3. **Rehearse** each candidate checkpoint 200–300 times.
4. **Calibrate** the confidence and set the help threshold.
5. **Deploy**, and compare the failures caught with the prediction of the twin.

What a user gets, and the gaps

Before the first real trial

- Which checkpoint to run in this workspace.
- The failures to expect, as rollout videos.
- The help threshold.

Prior work

SIMPLER and ReVeal rank policies in simulated and reconstructed scenes as real ones do. RialTo builds twins from quick scans.

ReVeal does not recover mass or friction. RialTo needs a person to set objects and physics, and it uses point clouds.

Pilot first, then the transfer test

Pilot (go/no-go)

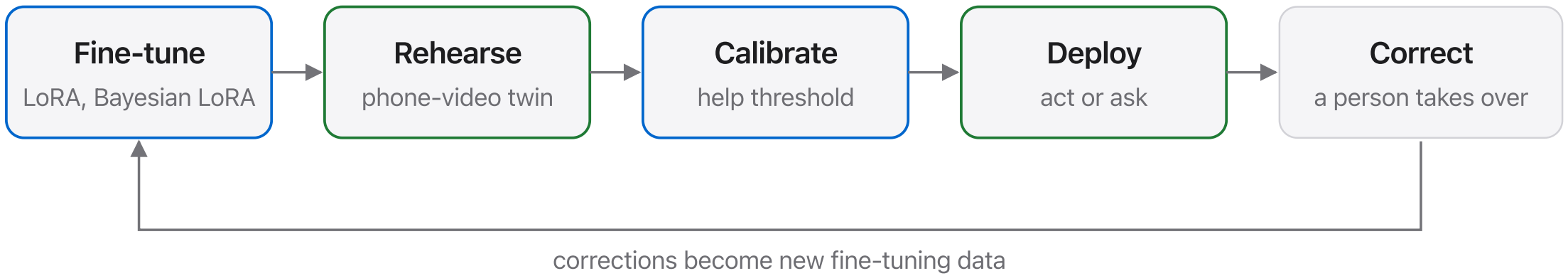
One arm, 2–3 rigid-object tasks, 3 checkpoints × 30 trials in the twin and on the robot. The work continues only if the twin ranks them in the real order and is within ~15 points.

Transfer test

Calibrators from (a) LIBERO, (b) the twin and (c) a few real rollouts are frozen with the twin threshold. Unseen real rollouts then score them.

The measures are the calibration error and, at a fixed help budget, caught failures, false alarms, missed failures and warning time.

How the two parts connect



■ research project: fine-tuning and calibration

■ application: twin and deployment

What this builds on

Work	What it does	What this proposal adds
Zollo and Zemel, 2025	Calibration of released VLAs on LIBERO, action-wise Platt scaling.	Varying the fine-tuning.
INSIGHT, RECALL	Help triggers and recovery data from token uncertainty.	Calibration of that uncertainty.
ReconVLA, VLAConf	Conformal and representation-level confidence.	Labeled rollouts from a twin.
Laplace-LoRA, LoRA ensembles	Post-hoc posterior or adapter ensemble. Less overconfident LLMs.	A test on a robot policy.
SIMPLER, ReVeal, RialTo	Evaluation in simulation and in twins from scans.	Calibration per workspace.

Sources

1. Kim et al. OpenVLA, arXiv:2406.09246: §5.2–5.3, Table 1 and footnote 4, Appendix E (LIBERO).
2. Zollo and Zemel. Confidence Calibration in VLA Models, 2025.
3. Karli et al. INSIGHT, 2025. Karli and Fitzgerald. RECALL, 2026.
4. Chen et al. ReconVLA, 2026. Huang et al. VLAConf, 2026.
5. Yang et al. Laplace-LoRA, 2023. Wang et al. LoRA ensembles, 2023.
6. Liu et al. LIBERO, 2023. LIBERO-Plus, 2025.
7. Li et al. SIMPLER, 2024. Wang et al. ReVeal, 2026.
8. Torne et al. RialTo, 2024.
9. This page: bushuyeu.com/review-openvla-visionary.